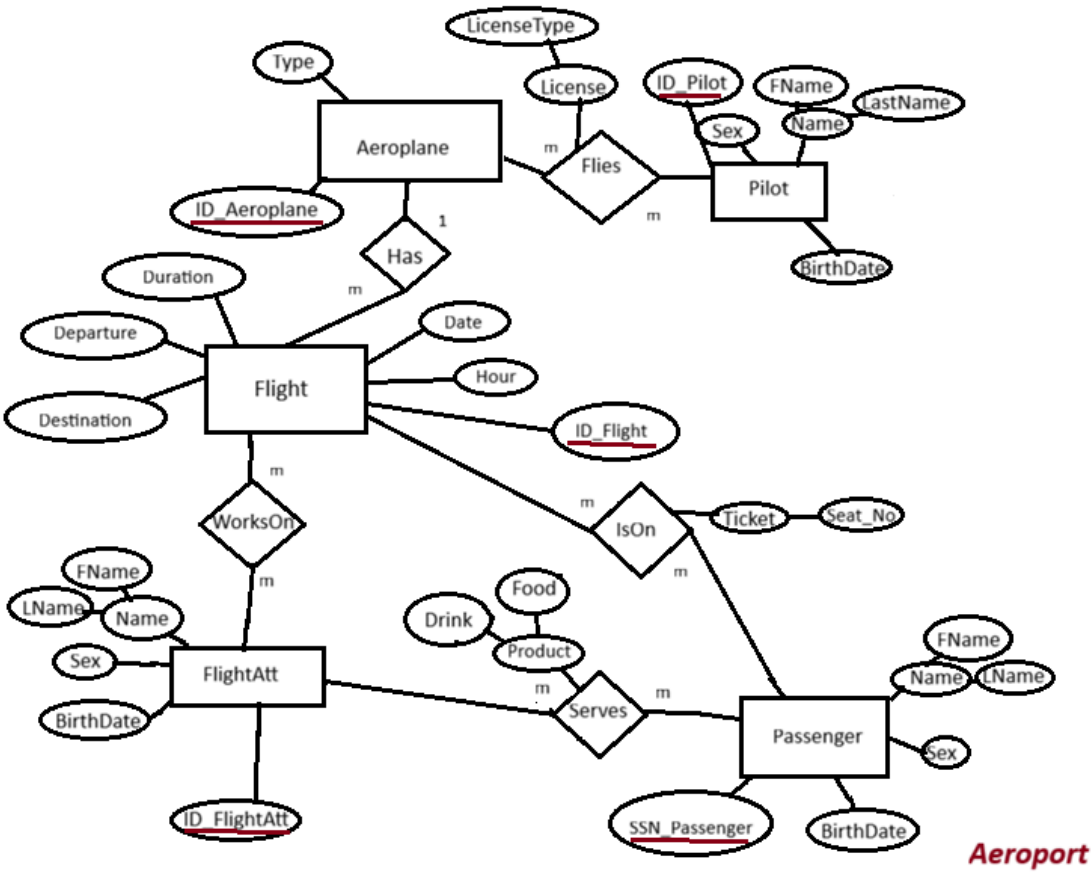
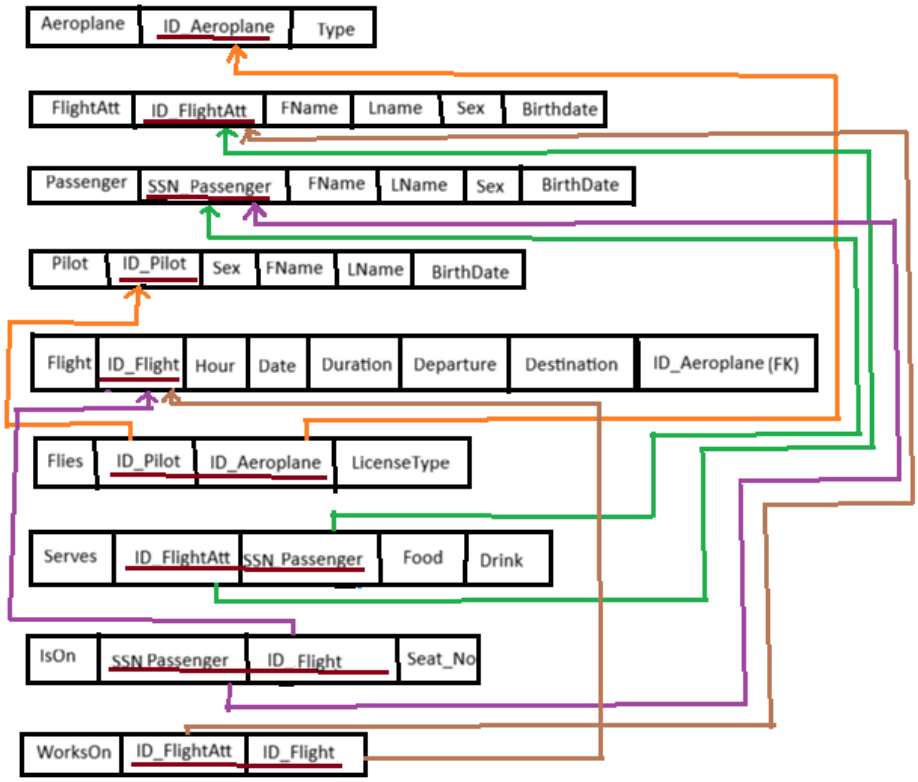
ProjectDB


LicenseType
Type
License
ID_Pilot
FName
LastName
Name
Sex
Aeroplane
m
Flies
Pilot
m
ID_Aeroplane
1
Has
BirthDate
Duration
m
Date
Departure
Flight
Hour
Destination
ID_Flight
m
m
WorksOn
IsOn
Ticket
Seat_No
m
FName
Food
LName
Name
Drink
Product
FName
Sex
FlightAtt
m
m
Name
LName
Serves
BirthDate
Passenger
Sex
ID_FlightAtt
SSN_Passenger
BirthDate
Aeroport

Mateescu Corina-Elena CEN3.2A



Relational Model

Table constraints:

```sql
CREATE TABLE tblPilot(
    ID_Pilot           char(8) PRIMARY KEY CLUSTERED,
    LastName     varchar(40) NOT NULL,
    FirstName  varchar(40) NOT NULL,
    Sex             char(1) CHECK (Sex IN ('M','F')) DEFAULT('M'),
    BirthDate     datetime
)
GO
CREATE TABLE tblPassenger(
    SSN_Passenger     char(8) PRIMARY KEY CLUSTERED,
    LastName     varchar(40) NOT NULL,
    FirstName  varchar(40) NOT NULL,
    Sex             char(1) CHECK (Sex IN ('M','F')) DEFAULT('M'),
    BirthDate     datetime
    )
GO
CREATE TABLE tblFlightAtt(
    ID_FlightAtt          char(8) PRIMARY KEY CLUSTERED,
    LastName     varchar(40) NOT NULL,
    FirstName  varchar(40) NOT NULL,
    Sex             char(1) CHECK (Sex IN ('M','F')) DEFAULT('M'),
    BirthDate     datetime
)
GO
```

```sql
CREATE TABLE tblAeroplane(
    ID_Aeroplane          char(8) PRIMARY KEY CLUSTERED,
    Type             char(10) CHECK (Type IN ('Private','Commercial')) DEFAULT('Commercial')
)
GO
CREATE TABLE tblFlight(
    ID_Flight        char(8) PRIMARY KEY CLUSTERED,
    ID_Aeroplane char(8) NOT NULL REFERENCES tblAeroplane(ID_Aeroplane)
        ON DELETE CASCADE ON UPDATE CASCADE,
    Date            date,
    Hour            time,
    Duration        varchar(5),
    Departure       varchar(40),
    Destination     varchar(40)
)
GO
```

```sql
CREATE TABLE tblServes(

    ID_FlightAtt char(8) NOT NULL REFERENCES tblFlightAtt(ID_FlightAtt)
        ON DELETE CASCADE ON UPDATE CASCADE,
    ID_Passenger char(8) NOT NULL REFERENCES tblPassenger(SSN_Passenger)
        ON DELETE CASCADE ON UPDATE CASCADE,
    Food varchar (30),
    Drink varchar (30),
    PRIMARY KEY (ID_FlightAtt, ID_Passenger)
    )

GO
CREATE TABLE tblWorksOn(

    ID_FlightAtt char(8) NOT NULL REFERENCES tblFlightAtt(ID_FlightAtt)
        ON DELETE CASCADE ON UPDATE CASCADE,
    ID_Flight char(8) NOT NULL REFERENCES tblFlight(ID_Flight)
        ON DELETE CASCADE ON UPDATE CASCADE,
    PRIMARY KEY (ID_FlightAtt, ID_Flight)
    )

GO
 CREATE TABLE tblFlies(
     ID_Pilot char(8) NOT NULL REFERENCES tblPilot(ID_Pilot)
         ON DELETE CASCADE ON UPDATE CASCADE,
     ID_Aeroplane char(8) NOT NULL REFERENCES tblAeroplane(ID_Aeroplane)
         ON DELETE CASCADE ON UPDATE CASCADE,
     LicenseType varchar (1) NOT NULL,
     PRIMARY KEY (ID_Pilot, ID_Aeroplane)
     )

 GO
 CREATE TABLE tblIsOn(
     ID_Passenger char(8) NOT NULL REFERENCES tblPassenger(SSN_Passenger)
         ON DELETE CASCADE ON UPDATE CASCADE,
     ID_Flight char(8) NOT NULL REFERENCES tblFlight(ID_Flight)
         ON DELETE CASCADE ON UPDATE CASCADE,
     Ticket varchar (10) NOT NULL,
     Seat_No  char(3),
     PRIMARY KEY (ID_Passenger, ID_Flight)
     )

 GO
```

Table content:

	ID_Pilot	LastName	FirstName	Sex	BirthDate
1	P00001	Popescu	Ion	M	1980-03-12 00:00:00.000
2	P00002	Ionescu	Ana	F	1985-07-24 00:00:00.000
3	P00003	Dumitrescu	Vlad	M	1978-11-02 00:00:00.000
4	P00004	Marinescu	Elena	F	1982-05-18 00:00:00.000
5	P00005	Georgescu	Mihai	M	1987-09-30 00:00:00.000

	SSN_Passenger	LastName	FirstName	Sex	BirthDate
1	PA00001	Popa	Dan	M	1995-04-10 00:00:00.000
2	PA00002	Ionescu	Elena	F	1996-08-22 00:00:00.000
3	PA00003	Dumitres...	Ioana	F	1994-02-14 00:00:00.000
4	PA00004	Marinescu	Vlad	M	1993-09-30 00:00:00.000
5	PA00005	Georges...	Ana	F	1992-12-11 00:00:00.000
6	PA00006	Stan	Mihai	M	1997-05-05 00:00:00.000
7	PA00007	Dobre	Radu	M	1998-01-21 00:00:00.000
8	PA00008	Vasilescu	Maria	F	1995-11-17 00:00:00.000

	ID_FlightAtt	LastName	FirstName	Sex	BirthDate
1	FA0001	Dumitrescu	Ioana	F	1993-02-14 00:00:00.000
2	FA0002	Vasilescu	Mihai	M	1994-05-20 00:00:00.000
3	FA0003	Popa	Elena	F	1992-08-08 00:00:00.000
4	FA0004	Ionescu	Andrei	M	1991-12-12 00:00:00.000

	ID_Aeroplane	Type
1	A0001	Private
2	A0002	Commercial
3	A0003	Commercial

	ID_Flight	ID_Aeroplane	Date	Hour	Duration	Departure	Destination
1	F0001	A0001	2025-01-10	08:30:00.0000000	90	Bucharest	Paris
2	F0002	A0002	2025-01-11	12:00:00.0000000	120	London	Rome
3	F0003	A0003	2025-01-12	18:45:00.0000000	150	Berlin	Madrid

	ID_Pilot	ID_Aeroplane	LicenseType
1	P00001	A0001	A
2	P00002	A0002	B
3	P00003	A0003	C
4	P00004	A0002	B
5	P00005	A0003	C

	ID_FlightAtt	ID_Passenger	Food	Drink
1	FA0001	PA00001	Sandwich	Water
2	FA0001	PA00003	Salad	Juice
3	FA0002	PA00002	Pasta	Water
4	FA0002	PA00004	Burger	Soda
5	FA0003	PA00005	Pizza	Juice
6	FA0003	PA00006	Salad	Water
7	FA0004	PA00007	Sandwich	Soda
8	FA0004	PA00008	Pasta	Juice

	ID_FlightAtt	ID_Flight
1	FA0001	F0002
2	FA0001	F0003
3	FA0002	F0001
4	FA0002	F0002
5	FA0003	F0001
6	FA0003	F0003
7	FA0004	F0002
8	FA0004	F0003

	ID_Passenger	ID_Flight	Ticket	Seat_No
3	PA00003	F0003	T003	3C
4	PA00004	F0002	T004	4A
5	PA00005	F0001	T005	5B
6	PA00006	F0003	T006	6C
7	PA00007	F0002	T007	7A
8	PA00008	F0001	T008	8B
9	PA00009	F0003	T009	9C
10	PA00010	F0002	T010	10A

Descriptions of the interrogations and their result:

```
--passengers with the food and drink they were served, and by which flight attendant--
SELECT p.SSN_Passenger, p.LastName AS PassengerLast, p.FirstName AS PassengerFirst,
       s.Food, s.Drink, fa.LastName AS FA_LastName, fa.FirstName AS FA_FirstName
FROM tblPassenger p
JOIN tblServes s ON p.SSN_Passenger = s.ID_Passenger
JOIN tblFlightAtt fa ON s.ID_FlightAtt = fa.ID_FlightAtt
ORDER BY fa.LastName, p.LastName;
```

	SSN_Passenger	PassengerLast	PassengerFirst	Food	Drink	FA_LastName	FA_FirstName
1	PA00003	Dumitrescu	Ioana	Salad	Juice	Dumitrescu	Ioana
2	PA00001	Popa	Dan	Sandwich	Water	Dumitrescu	Ioana
3	PA00007	Dobre	Radu	Sandwich	Soda	Ionescu	Andrei
4	PA00008	Vasilescu	Maria	Pasta	Juice	Ionescu	Andrei
5	PA00005	Georgescu	Ana	Pizza	Juice	Popa	Elena
6	PA00006	Stan	Mihai	Salad	Water	Popa	Elena
7	PA00002	Ionescu	Elena	Pasta	Water	Vasilescu	Mihai
8	PA00004	Marinescu	Vlad	Burger	Soda	Vasilescu	Mihai

```
--flight attendants and the flights they work on--
SELECT fa.ID_FlightAtt,
       fa.LastName,
       fa.FirstName,
       w.ID_Flight
FROM tblFlightAtt fa
JOIN tblWorksOn w
     ON fa.ID_FlightAtt = w.ID_FlightAtt
ORDER BY fa.LastName, w.ID_Flight;
```

	ID_FlightAtt	LastName	FirstName	ID_Flight
1	FA0001	Dumitrescu	Ioana	F0002
2	FA0001	Dumitrescu	Ioana	F0003
3	FA0004	Ionescu	Andrei	F0002
4	FA0004	Ionescu	Andrei	F0003
5	FA0003	Popa	Elena	F0001
6	FA0003	Popa	Elena	F0003
7	FA0002	Vasilescu	Mihai	F0001
8	FA0002	Vasilescu	Mihai	F0002

```sql
--commercial aeroplanes with pilots who are licensed to fly them--
SELECT a.ID_Aeroplane, a.Type, pi.ID_Pilot, pi.LastName, f.LicenseType
FROM tblAeroplane a
JOIN tblFlies f ON a.ID_Aeroplane = f.ID_Aeroplane
JOIN tblPilot pi ON f.ID_Pilot = pi.ID_Pilot
WHERE a.Type = 'Commercial'
ORDER BY pi.LastName, f.LicenseType;
```

	ID_Aeroplane	Type	ID_Pilot	LastName	LicenseType
1	A0003	Commercial	P00003	Dumitrescu	C
2	A0003	Commercial	P00005	Georgescu	C
3	A0002	Commercial	P00002	Ionescu	B
4	A0002	Commercial	P00004	Marinescu	B

```sql
--passengers per flight attendant for flights with more than 1 passenger--
SELECT fa.ID_FlightAtt, fa.LastName AS FA_LastName, COUNT(s.ID_Passenger) AS NumPassengers
FROM tblFlightAtt fa
JOIN tblServes s ON fa.ID_FlightAtt = s.ID_FlightAtt
GROUP BY fa.ID_FlightAtt, fa.LastName
HAVING COUNT(s.ID_Passenger) > 1
ORDER BY NumPassengers DESC;
```

	ID_FlightAtt	FA_LastName	NumPassengers
1	FA0001	Dumitrescu	2
2	FA0002	Vasilescu	2
3	FA0003	Popa	2
4	FA0004	Ionescu	2

```sql
--number of times flight attendants worked togheter--
SELECT w1.ID_FlightAtt AS FA1,
       w2.ID_FlightAtt AS FA2,
       COUNT(*) AS TimesTogether
FROM tblWorksOn w1
JOIN tblWorksOn w2
     ON w1.ID_Flight = w2.ID_Flight
    AND w1.ID_FlightAtt < w2.ID_FlightAtt
GROUP BY w1.ID_FlightAtt, w2.ID_FlightAtt
ORDER BY TimesTogether DESC;
```

	FA1	FA2	TimesTogether
1	FA0001	FA0004	2
2	FA0002	FA0004	1
3	FA0003	FA0004	1
4	FA0001	FA0002	1
5	FA0001	FA0003	1
6	FA0002	FA0003	1

```sql
--youngest passenger on each aeroplane--
SELECT f.ID_Aeroplane,
       a.Type,
       MAX(p.BirthDate) AS YoungestBirthDate
FROM tblIsOn i
JOIN tblPassenger p
     ON i.ID_Passenger = p.SSN_Passenger
JOIN tblFlight f
     ON i.ID_Flight = f.ID_Flight
JOIN tblAeroplane a
     ON f.ID_Aeroplane = a.ID_Aeroplane
GROUP BY f.ID_Aeroplane, a.Type
ORDER BY YoungestBirthDate DESC;
```

	ID_Aeroplane	Type	YoungestBirthDate
1	A0002	Commercial	1998-01-21 00:00:00.000
2	A0003	Commercial	1997-05-05 00:00:00.000
3	A0001	Private	1996-08-22 00:00:00.000

```sql
--license types used by only one pilot--
SELECT LicenseType,
       COUNT(DISTINCT ID_Pilot) AS NumPilots
FROM tblFlies
GROUP BY LicenseType
HAVING COUNT(DISTINCT ID_Pilot) = 1;
```

	LicenseType	NumPilots
1	A	1

```sql
--passengers served by each flight attendant per aeroplane--
SELECT fa.ID_FlightAtt,
       fa.LastName AS FA_LastName,
       f.ID_Aeroplane,
       COUNT(DISTINCT s.ID_Passenger) AS NumPassengers
FROM tblFlightAtt fa
JOIN tblServes s
     ON fa.ID_FlightAtt = s.ID_FlightAtt
JOIN tblWorksOn w
     ON fa.ID_FlightAtt = w.ID_FlightAtt
JOIN tblFlight f
     ON w.ID_Flight = f.ID_Flight
GROUP BY fa.ID_FlightAtt, fa.LastName, f.ID_Aeroplane
ORDER BY NumPassengers DESC, fa.LastName;
```

	ID_FlightAtt	FA_LastName	ID_Aeroplane	NumPassengers
1	FA0001	Dumitrescu	A0002	2
2	FA0001	Dumitrescu	A0003	2
3	FA0004	Ionescu	A0002	2
4	FA0004	Ionescu	A0003	2
5	FA0003	Popa	A0003	2
6	FA0003	Popa	A0001	2
7	FA0002	Vasilescu	A0001	2
8	FA0002	Vasilescu	A0002	2

```sql
--aeroplanes with all pilots licensed to fly them--
SELECT a.ID_Aeroplane, a.Type, pi.LastName AS PilotLast, f.LicenseType
FROM tblAeroplane a
JOIN tblFlies f ON a.ID_Aeroplane = f.ID_Aeroplane
JOIN tblPilot pi ON f.ID_Pilot = pi.ID_Pilot
WHERE f.LicenseType IN ('A','B','C')
ORDER BY a.Type, pi.LastName;
```

	ID_Aeroplane	Type	PilotLast	LicenseType
1	A0003	Commercial	Dumitrescu	C
2	A0003	Commercial	Georgescu	C
3	A0002	Commercial	Ionescu	B
4	A0002	Commercial	Marinescu	B
5	A0001	Private	Popescu	A

```sql
--passengers and the total number of different flight attendants they were served by--
SELECT p.SSN_Passenger,
       p.LastName,
       p.FirstName,
       COUNT(DISTINCT s.ID_FlightAtt) AS NumFlightAttendants
FROM tblPassenger p
JOIN tblServes s
     ON p.SSN_Passenger = s.ID_Passenger
GROUP BY p.SSN_Passenger, p.LastName, p.FirstName
ORDER BY NumFlightAttendants DESC, p.LastName;
```

	SSN_Passenger	LastName	FirstName	NumFlightAttendants
1	PA00007	Dobre	Radu	1
2	PA00003	Dumitrescu	Ioana	1
3	PA00005	Georgescu	Ana	1
4	PA00002	Ionescu	Elena	1
5	PA00004	Marinescu	Vlad	1
6	PA00001	Popa	Dan	1
7	PA00006	Stan	Mihai	1
8	PA00008	Vasilescu	Maria	1